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Option D – **Energy and the Environment**

9. *(a)* (i) The total power emitted by the Sun is 4.0×10^{26} W. Given that the Earth is 1.5×10^{11} m from the Sun show that the intensity of the radiation reaching the Earth's outer atmosphere is approximately $1\,400\,\text{W m}^{-2}$. [2]

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(ii) The mean annual power available at the Earth's surface in the UK is only about 20% of the answer to part *(a)*(i). By the middle of 2015 solar photovoltaic (PV) installations accounted for approximately 8 GW of the UK's electricity production. Given that solar panels are only 15% efficient at converting solar radiation into electricity, estimate the total area of solar panels in the UK in 2015. [2]

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(iii) Give a reason why the amount of electrical energy produced by a solar farm in Portugal is likely to be greater than that from a solar farm of the same size in the UK per year. [1]

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(b) (i) State Wien's law. [1]

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(ii) Assuming a mean temperature of 290 K for the Earth and that it emits radiation from its surface as a black body, show that the peak wavelength of the radiation emitted by the Earth is in the infra-red region of the electromagnetic spectrum. [2]

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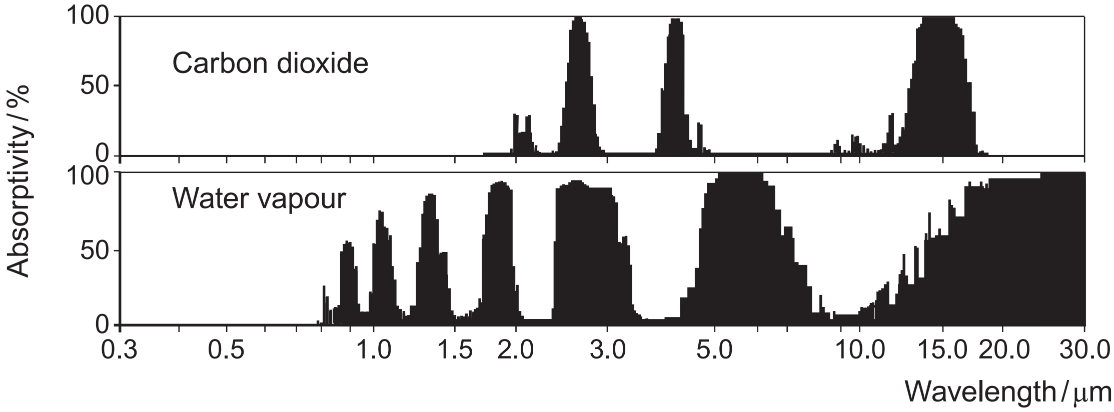
(iii) The graphs below show greenhouse gas absorption spectra for water vapour and carbon dioxide as a function of the wavelength of the radiation incident on the gas. An absorptivity of zero % means no radiation is absorbed, whilst an absorptivity of 100% means that all the incident radiation is absorbed.



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I. Use the graphs to explain how increased levels of water vapour and carbon dioxide in the atmosphere can account for the greenhouse effect. [4]

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II. The term *'positive feedback'* is used to describe a system's output influencing the system so as to increase the output even further. The Earth's surface and atmosphere may be regarded as a system, and temperature as the system's output. Explain how the greenhouse effect is part of a positive feedback. [2]

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(c) (i) Define the term U-value and state **one** factor which affects the U-value of a material. [2]

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(ii) A room in a large office block is maintained at a constant temperature of 21 °C. The room has one of its walls, measuring 5.0 m × 3.0 m, exposed to the outside. This wall is double glazed (made of two layers of glass with a small air filled gap between them). Negligible heat is lost through the other walls of the room.
Calculate the energy per second transmitted through the outside wall if the outside temperature is 9 °C. [U-value of double glazing glass = 1.8 W m^{-2} K^{-1}] [2]

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(iii) The heating system in the room can provide a maximum output of 1 kW. Determine the lowest outside temperature for which the double glazing can maintain an internal temperature of 21 °C. [2]

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